1.3 Chemical Bonding

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Hard

Time allowed: 20

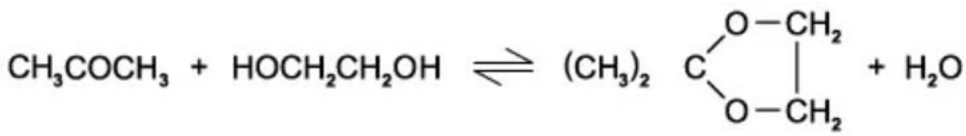
Score: /9

Percentage: /100

Question 1

1.00g of propanone is reacted with 5.00g of ethane-1,2-diol and the strong acid benzene sulphonic acid, $C_6H_5SO_3H$. The mixture is heated under reflux in an inert solvent.

Y



Which of the following statements is not correct?

A. Propanone has a higher melting point than ethane-1,2-diol

B. Both products are polar molecules

C. Ethane-1,2-diol is soluble in water

D. Water has a higher melting point than liquid Y

[1 mark]

Question 2

Hydrogen bonding occurs between molecules of liquid methanal, HCHO, and molecules of liquid Y. Which of the following is most likely to be liquid Y?

A. $CH_3CO_2CH_3$

B. CH_3OH

C. CH_3COCH_3

D. CH_3CHO

[1 mark]

Question 3

A student is studying conversions of different species. The two conversions below show similarities.

$$PCl_5 \rightarrow PCl_6^-$$

$$C_2H_4 \rightarrow C_2H_6$$

What similar features do these two conversions have?

A. Decrease in bond angle of the species involved

B. Disappearance of a π bond

C. Change in oxidation state of an element

D. A lone pair of electrons in the product

[1 mark]

Question 4

Equal amounts of 1–chloroethane, 1–bromoethane and 1–iodoethane were added to separate test-tubes each containing 2.0 cm^3 of aqueous silver nitrate. The test tubes were heated and a cloudy precipitate formed in each tube.

The equation below, within which X represents a halogen atom, and R represents CH_3CH_2, describes the hydrolysis reaction occurring within the test tubes.

$$H_2O(l) + R\text{-}X(l) + Ag^+(aq) \quad CH_3CH_2\text{-}OH(aq) + AgX(s) + H^+(aq)$$

The rate of AgX production, as measured by the speed of precipitation formation, was in the order AgI > AgBr > AgCl.

Why is this?

A. The first ionisation energy of the halogen decreases from Cl to I

B. The solubility of AgX(s) decreases from AgCl to AgI

C. The bond energy of R–X decreases from RCl to RI

D. The R–X bond polarity decreases from RCl to RI

[1 mark]

Question 5

Concerning their Pauling electronegativity values, which atom is more likely to form a **covalent** bond with fluorine?

	Atom	**Pauling value**
	Fluorine	4.0
A	Hydrogen	2.2
B	Copper	1.95
C	Magnesium	1.3
D	Potassium	0.82

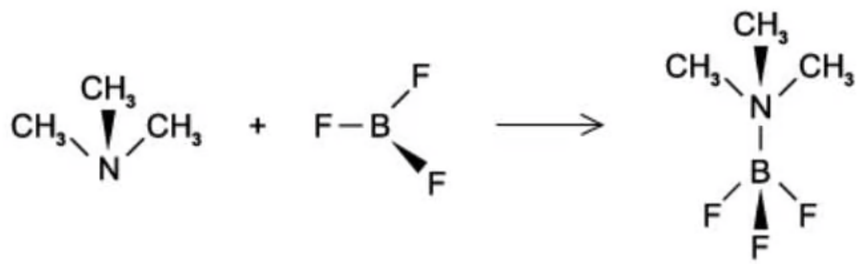
[1 mark]

Question 6

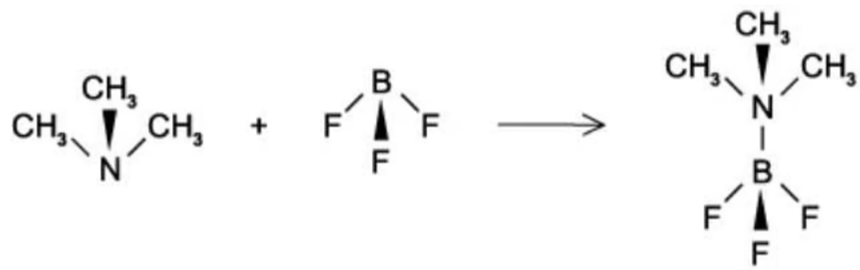
Boron trifluoride, BF_3, reacts with Trimethylamine, $(CH_3)_3N$, to form a compound of formula $(CH_3)_3N.BF_3$.

How may this reaction be written in terms of the shapes of the reactants and products?

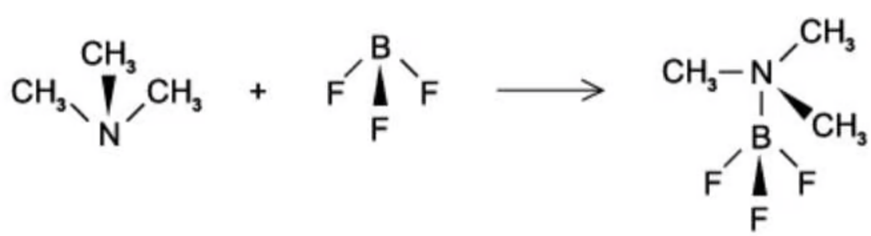
A.



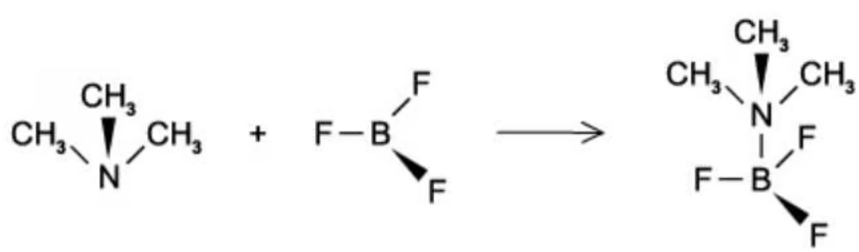
B.



C.



D.



[1 mark]

Question 7

$CHCl_3$ and Br_2 are both liquids at room temperature due to the existence of dipoles.

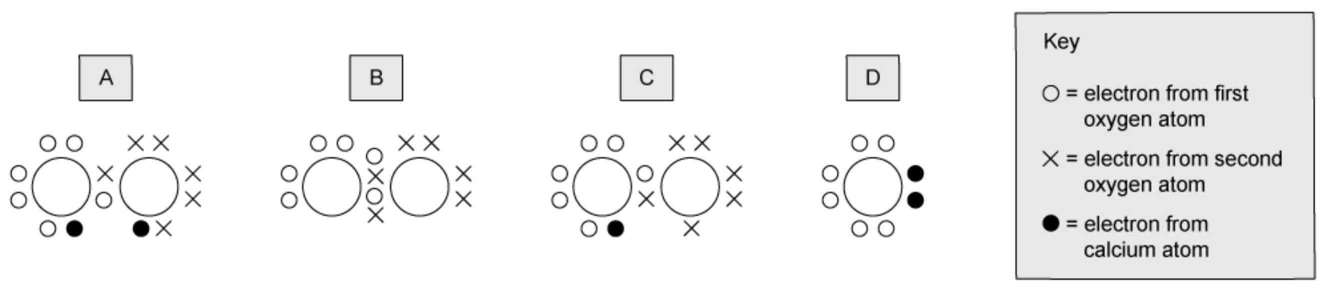
What dipoles are involved within $CHCl_3$ and Br_2?

	$CHCl_3$	Br_2
A	Induced dipoles only	Induced dipoles only
B	Induced dipoles only	Induced dipoles and permanent dipoles
C	Induced dipoles and permanent dipoles	Induced dipoles and permanent dipoles
D	Induced dipoles and permanent dipoles	Induced dipoles only

[1 mark]

Question 8

Calcium Oxide, CaO_2, can be used in toothpaste as a teeth whitener. It is formed when calcium metal burns in oxygen. Which dot-and-cross diagram correctly illustrates the electron structure of the peroxide ion in CaO_2.



[1 mark]

Question 9

Bones contain a molecule called hydroxyapatite, $Ca(OH)_2.3Ca_3(PO_4)_2$, which functions as a strengthening agent.

As you age, calcium ions can be lost from hydroxyapatite, causing the bones to weaken. Strontium salt tablets can be taken to strengthen bones as the strontium ions replace the calcium ions lost from hydroxyapatite

Which of these is a correct statement?

A. Strontium hydroxide is less soluble than calcium hydroxide and so will precipitate better in the bone structure.

B. There is ionic, covalent and metallic bonding in hydroxyapatite which gives it strength.

C. Strontium ions are nearly the same size as calcium ions and so may easily replace them in the hydroxyapatite.

D. Strontium forms a covalent bond with phosphate.

[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- Propanone is a polar molecule and has dipole-dipole interactions between the molecules.
- Ethane-1,2–diol has hydrogen bonds between the molecules due to the two hydroxide groups creating δ^+ hydrogen atoms that will form a hydrogen bond with the oxygen.
- Hydrogen bonding is stronger than dipole-dipole interactions so ethane-1,2–diol will have a higher melting point than propanone.

B is incorrect as both products are polar molecules.

C is incorrect as both liquid Y and water will have polar regions.

D is incorrect as water does have a higher melting point than liquid Y.

2

The correct answer is **B** because:

- A hydrogen bond is an intermolecular interaction between a hydrogen attached to a highly electronegative element (creating a δ^+ hydrogen), with an atom with a lone pair in another molecule.
- The hydroxyl group can easily form hydrogen bonds with another molecule.
- Methanol has a hydroxyl group so is most likely to be liquid Y.

A is incorrect as methyl ethanoate contains no OH group, so there is no hydrogen bonding.

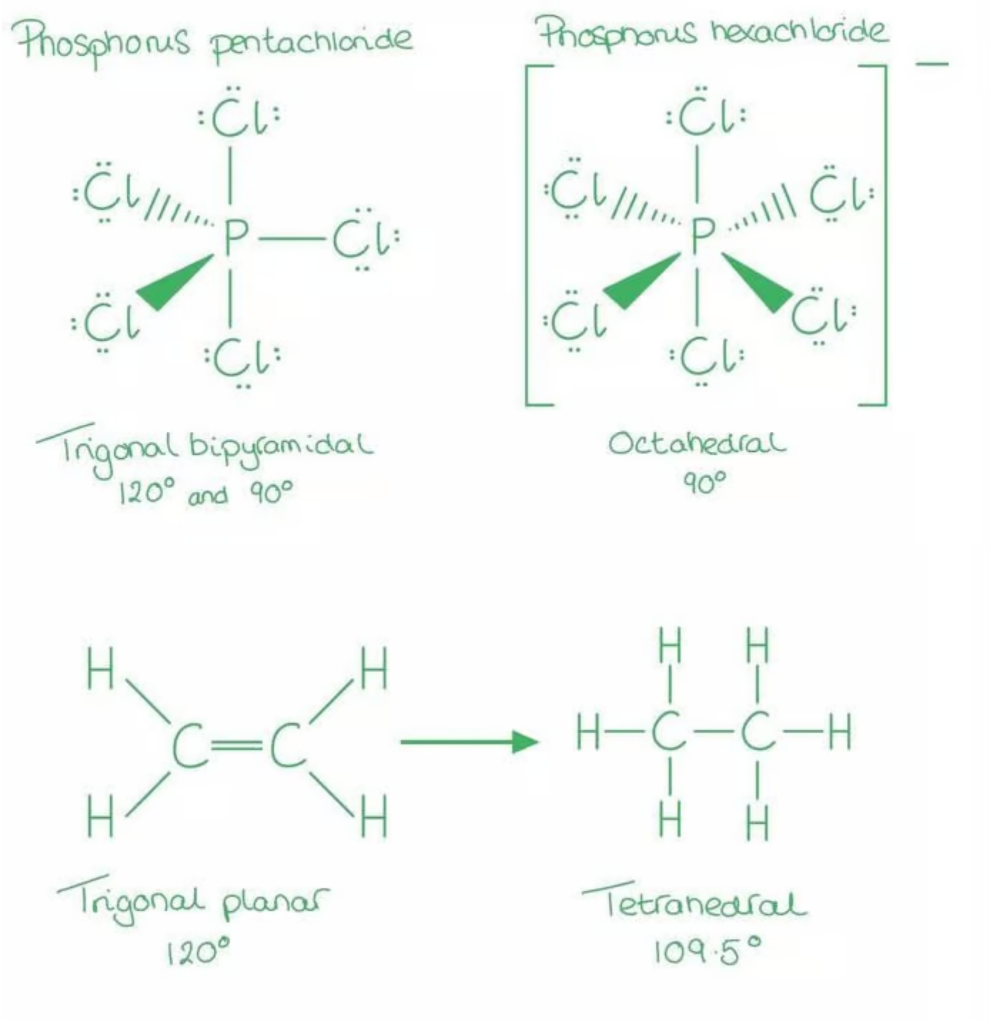
C is incorrect as propanone will not form hydrogen bonds as there are no hydrogen atoms attached to the oxygen.

D is incorrect as ethanal will not form hydrogen bonds as there are no hydrogen atoms attached to the oxygen.

3

The correct answer is **A** because:

- The **VESPR theory** states that electron pairs **repel** each other whether or not they are in bond pairs or lone pairs; therefore the electrons will spread as much as possible to **reduce repulsion.**
- The bond angle is affected by the presence of lone pair of electrons at the central atom.
  - Each lone pair reduces the original angle (if there were no lone pairs) by about 2.5˚



B is incorrect as the formation of PCl_5 to PCl_6^- will not result in the disappearance of a π bond. The formation of C_2H_6 will result in the removal of a π bond as the double bond is broken to a single bond.

C is incorrect as the oxidation state of phosphorus is not changing in $PCl_5 \rightarrow PCl_6^-$.

D is incorrect as there are no lone pairs of electrons in C_2H_4 or C_2H_6.

4

The correct answer is **C** because:

- The atomic radius of the halogens increases as the periods increase down the group.
  - The atomic radius is governed by; the number of shells, the force between the nucleus and the outer shell electrons.
- The electronegativity of the group decreases down the group.
- The bond energy in the R-X molecules will decrease down the group meaning that halogenoalkanes further down the group will more readily form new products.
- It is difficult to break a carbon-fluorine bond, but easy to break a carbon-iodine one.

A is incorrect as the first ionisation energy is the energy involved in removing one mole of electrons from one mole of gaseous atoms. This does not apply to this reaction.

B is incorrect as chlorine, bromine and iodine all dissolve in water to some extent, but there is no pattern to this and would not explain these results.

D is incorrect as bond polarity is the separation of electric charge along a bond, leading to a chemical having a dipole, fluorine forms a strong dipole whereas iodine is equally electronegative than carbon. While this may explain the bond strength it is the energy required to break the bond that will determine the rate of the reaction.

5

The correct answer is **A** because:

- Electronegativity is a measure of the tendency of an atom to attract a bonding pair of electrons.

- When two atoms of equal electronegativity bond together they both have the

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5

The correct answer is **A** because:

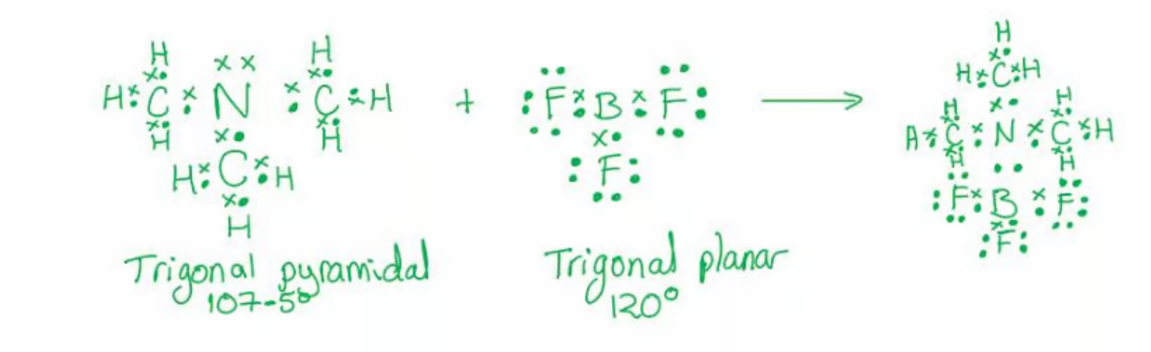
- Electronegativity is a measure of the tendency of an atom to attract a bonding pair of electrons.
- When two atoms of equal electronegativity bond together they both have the same tendency to attract the bonding pair of electrons, so it will be found on average halfway between the two atoms.
  - This will produce a 'pure covalent bond', (a difference of less than 0.5, ΔEN, between the electronegativities)
- If one of the atoms is more electronegative than the other (as in this case), the electrons in the bond will be more attracted to the more electronegative atom, creating a polar bond.
  - This describes most covalent bonds (a difference between the electronegativities of atoms, ΔEN, of 0.5 and 1.6)

If the difference between the electronegativities of the atoms is greater than 2.0, ΔEN, the bond is considered to be ionic.

6

The correct answer is **A** because:

- The **VESPR theory** states that electron pairs **repel** each other whether or not they are in bond pairs or lone pairs; therefore the electrons will spread as much as possible to **reduce repulsion.**
    - This creates the shape of the molecules based on the atoms and lone pairs present.
- Trimethylamine has a **lone pair of electrons** on the nitrogen atom, giving it the structure of **trigonal pyramidal** (107.5°).
- Boron trifluoride has no lone pair of electrons forming a **trigonal planar** molecule with angles of 120°.
- The molecule formed in this reaction will have a **dative covalent bond** between the two structures as boron is described as **electron-deficient.**



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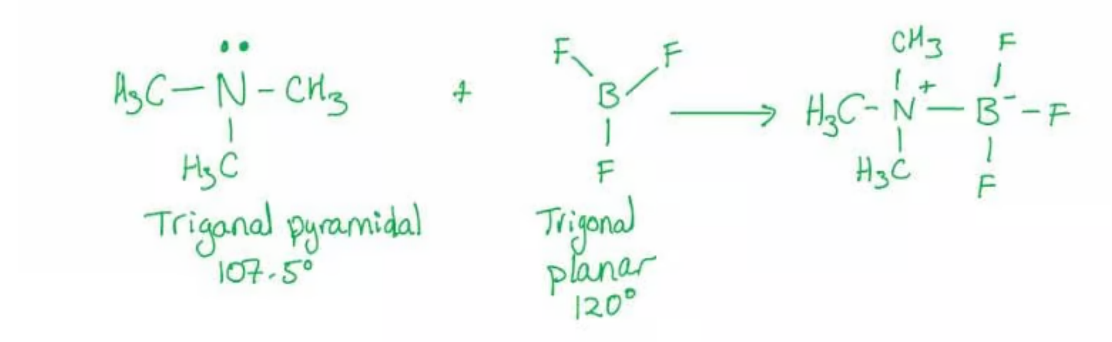

B & **C** are incorrect as they do not show the correct structure of boron trifluoride, it is drawn as if a lone pair of electrons are present.

D is incorrect as this structure is not taking into account the repulsion from the dative bonding pair between the nitrogen and the boron.

7

The correct answer is **D** because:

- Electronegativity is a measure of the tendency of an atom to attracts a bonding pair of electrons.
- When two atoms of equal electronegativity bond together they both have the same tendency to attract the bonding pair of electrons, so it will be found on average halfway between the two atoms.
  - This will produce a 'pure covalent bond', (a difference of less than 0.5, ΔEN, between the electronegativities).
  - This will be the case between the bromine atoms as they have equal electronegativity.
  - Bromine will experience induced dipoles only when a polar molecule disturbs the arrangement of the electrons.
- If one of the atoms is more electronegative than the other, the electrons in the bond will be more attracted to the more electronegative atom, creating a polar bond.
  - This describes most covalent bonds (a difference between the electronegativities of atoms, ΔEN, of 0.5 and 1.6).
  - Chlorine is more electronegative than carbon creating a polar covalent bond between the chlorine atoms and the carbon atoms.
  - This is described as a permanent dipole.
  - The molecule will also experience induced dipoles when another polar molecule disturbs the electron arrangement of the electrons.

A & **B** are incorrect as chlorine is more electronegative than carbon so will have permanent dipoles.

C is incorrect as bromine will not have permanent dipoles as the electronegativity of the two atoms is equal.

8

    - The molecule will also experience induced dipoles when another polar molecule disturbs the electron arrangement of the electrons.

A & **B** are incorrect as chlorine is more electronegative than carbon so will have permanent dipoles.

C is incorrect as bromine will not have permanent dipoles as the electronegativity of the two atoms is equal.

8

The correct answer is **A** because:

- Calcium is in group 2 with 2 valence electrons, meaning the calcium ion has a 2^+ charge, Ca^{2+}.
- The peroxide ion is made up of two oxygen atoms with a charge of 2^-, O_2^{2-}.
- When they bond ionically, the calcium will give each oxygen atom one electron.
- This is shown correctly in structure **A.**

B is incorrect as this shows a double bond between the oxygen atoms and would not be the peroxide ion but O_2 molecule.

C is incorrect as the peroxide ion has a 2^- charge, and calcium has a 2^+ charge, so two electron pairs are used in the bond in calcium oxide.

D is incorrect as there is only one oxygen atom present.

9

The correct answer is **C** because:

- **Strontium** has a chemical similarity to **calcium** and will replace calcium in bone.
    - Both calcium and strontium are in group 2.
    - Both have two valence electrons.
- Strontium is more soluble in water so more likely to be absorbed into the body to replace the calcium ions lost in bone.

A is incorrect as strontium hydroxide is **more soluble** than calcium hydroxide. The

hydroxides of group 2 metals increase as the periods increase in the group.

B is incorrect as there are no **metallic bonds** present in hydroxyapatite; metallic